

Random Connection Hypergraphs

Peter Juhász



Department of Mathematics
Aarhus University



Danish Data Science Academy
Copenhagen, Denmark

November 13, 2024



Morten Brun



Christian Hirsch



Moritz Otto

M. Brun, C. Hirsch, P. Juhász, and M. Otto.
Random connection hypergraphs.
ArXiv preprint arxiv:2407.16334, 2024.

Table of Contents

1 Introduction

2 Model

3 Results

4 Proofs

Table of Contents

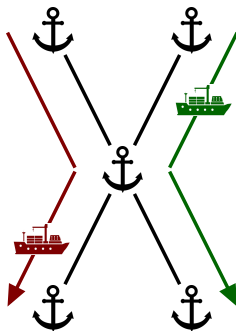
1 Introduction

2 Model

3 Results

4 Proofs

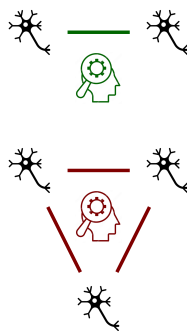
Complex Systems



(a) Maritime Traffic



(b) Collaborations



(c) Neural Network

Simple graphs: information loss

Outline

Goal: develop a stochastic network model
for real-world higher-order networks

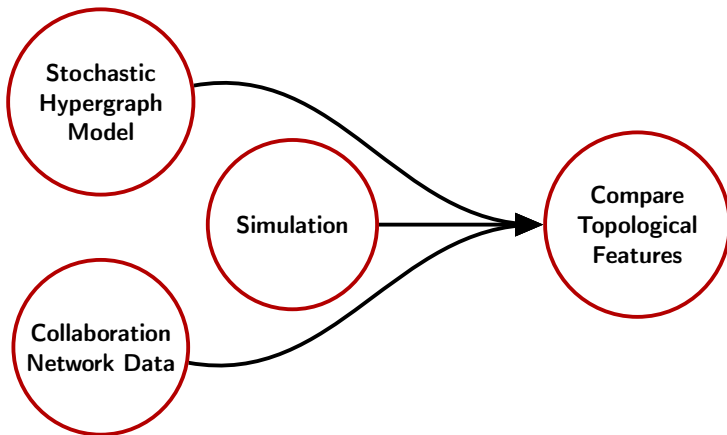


Table of Contents

1 Introduction

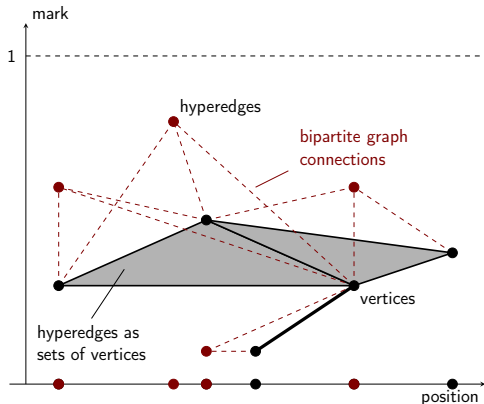
2 Model

3 Results

4 Proofs

Random Connection Hypergraph Model (RCHM)

- vertices: marked point process on a metric space
- **hyperedges**: marked point process on a metric space
- bipartite graph connections: connection function
- hypergraph formed of vertices



Random Connection Hypergraph Model (RCHM)

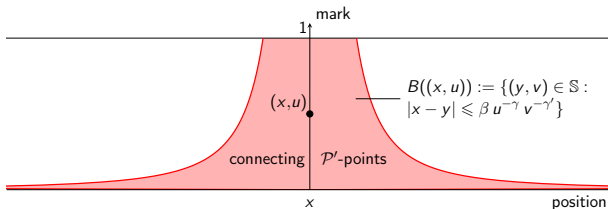
Two Poisson point processes
on $\mathbb{S} := \mathbb{R} \times [0, 1]$

- vertices: $\mathcal{P} \subset \mathbb{S}$
- hyperedges: $\mathcal{P}' \subset \mathbb{S}$

Connections:

$$\{(x, u), (y, v)\} \in \mathcal{P} \times \mathcal{P}'$$

$$|x - y| \leq \beta u^{-\gamma} v^{-\gamma'}$$



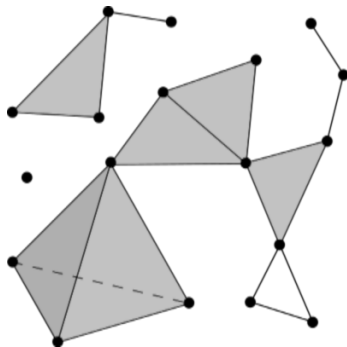
Motivation: Gracar, P., et al. "The age-dependent random connection model." *Queueing Systems* 93 (2019): 309-331.

Simplicial Complex

Hyperedges in $\mathcal{P}$ constitutes the $G(\mathcal{P}, \mathcal{P}')$ simplicial complex

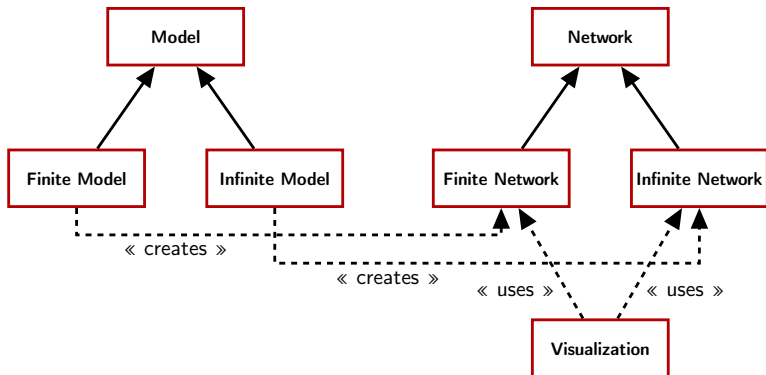
Simplicial Complex

- (P, Σ)
- $P \neq \emptyset, |P| < \infty$
- $\Sigma \subseteq \mathcal{P}(P)$
- $p \in P \implies \{p\} \in \Sigma$
- $\sigma \in \Sigma, \Delta \subset \sigma \implies \Delta \in \Sigma$



- Realization of RCHM: simplicial complex
- Simplices: $\Sigma = \{\Delta \subseteq \mathcal{P} : \exists p' \in \mathcal{P}' : \forall p \in \Delta, p \leftrightarrow p'\}$

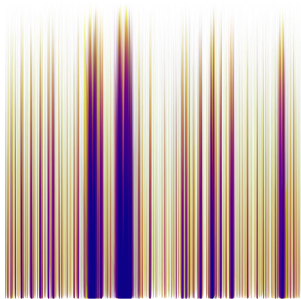
Simulation



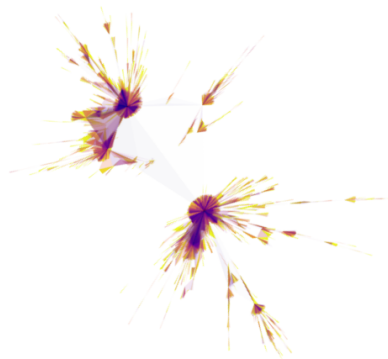
- Finite-Size Effects
- Conjectures

- **C++:** 4 149 SLoC
- **Python:** 6 654 SLoC

A Simulated Network



(a) Simulated Network



(b) Force-Directed Layout

Network Data

- Collaboration network data
- Source: arXiv preprint server
- Vertices: $\sim 8 \cdot 10^5$ authors
- Simplices: $\sim 10^6$ documents

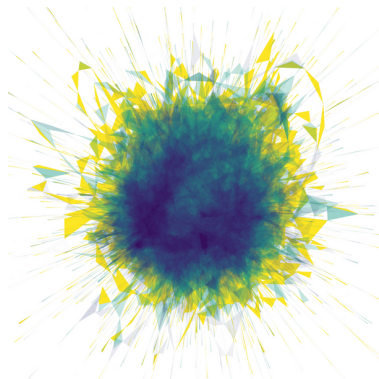


Table of Contents

1 Introduction

2 Model

3 Results

4 Proofs

Palm Distribution

$$\Sigma_m^A := \{\Delta_m \in \Sigma_m : c(\Delta_m) \in A\} \quad A \subset \mathbb{R}$$

Palm distribution:

$$\begin{aligned} & \mathbb{E}[f(\Delta_m^*, \mathcal{P}, \mathcal{P}')] \\ &= \frac{1}{\lambda_m} \mathbb{E}\left[\sum_{\Delta_m \in \Sigma_m^A} f(\Delta_m - c(\Delta_m), \mathcal{P} - c(\Delta_m), \mathcal{P}' - c(\Delta_m)) \right] \end{aligned}$$

Proposition (Finiteness of the m -simplex-intensity)

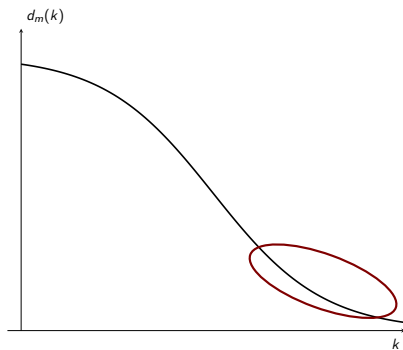
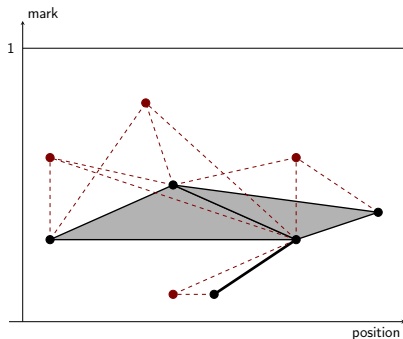
- $\mathcal{P}, \mathcal{P}'$ are independent Poisson point processes on $\mathbb{S}$
- $A \subset \mathbb{R}$ with $|A| = 1$
- $m \geq 0, \gamma < 1, \gamma' < 1/(m+1), \beta > 0$

Then $\lambda_m = \mathbb{E}[|\Sigma_m^A|] \in (0, \infty)$ independent of A

Higher-Order Degree Distribution

$$\deg(\Delta) := \mathcal{P}'\left(\bigcap_{p \in \Delta} B(p)\right)$$

$$d_m(k) := \mathbb{P}(\deg(\Delta_m^*) \geq k)$$



Higher-Order Degree Distributions

Theorem

If

$$d_m(k) = \mathbb{P}(\deg(\Delta_m^*) \geq k)$$

Then

$$\lim_{k \uparrow \infty} \log(d_m(k)) / \log(k) = m - (m+1)/\gamma$$

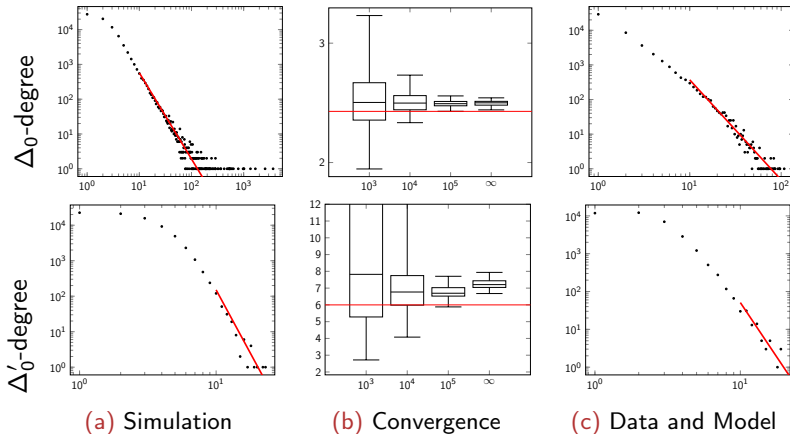
Proof Steps.

- Define Palm distribution
- Lower bound: find $B_k \subset \mathbb{S}$ in which points surely connect
- Upper bound: consider pairs of $\mathcal{P}$ -vertices in Δ



Parameter Fitting – Statistics Dataset

Remark: model is symmetric if $\mathcal{P}$ and $\mathcal{P}'$ -vertices are exchanged.



γ and γ' are determined from the fitted exponents

Asymptotics of Simplex Counts

- $S_n := \#\{\Delta_m \in \Sigma_m : c(\Delta_m) \in [0, n]\}$

Theorem

Central Limit Theorem

- $0 < \gamma < \frac{1}{2}; 0 < \gamma' < \frac{1}{2m+1}$
- $n^{-\frac{1}{2}}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Stable Limit Law

- $\frac{1}{2} < \gamma < 1; 0 < \gamma' < \frac{1}{2m+1}$
- $n^{-\gamma}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{S}(1/\gamma)$

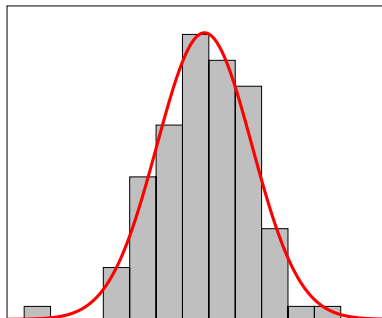
Proof Sketch.

- Harris-FKG theorem
- Upper bound sum of covariance terms
- Apply CLT (associated rvs)
- Neglect high-mark vertices
- Approximate low-mark vertices as iid Pareto rvs

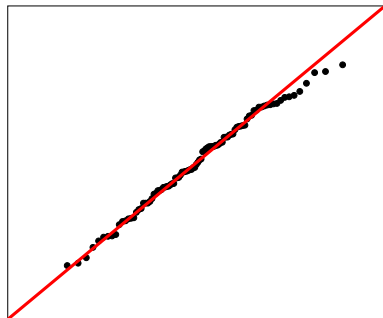


Normal Distribution of Triangle Counts

$\gamma = 0.25, \gamma' = 0.10$: Normal Distribution



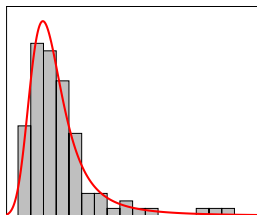
(a) Triangle Counts



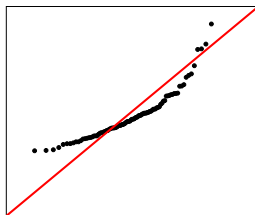
(b) Normal Distribution

Stable Distribution of Triangle Counts

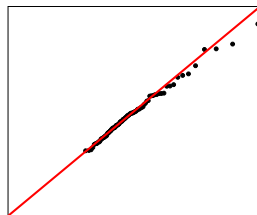
$\gamma = 0.75, \gamma' = 0.10$: Stable Distribution



(a) Triangle Counts

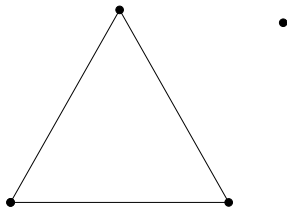


(b) Normal Distribution



(c) Stable Distribution

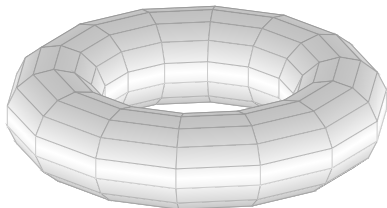
Betti Numbers



$$\beta_0 = 2$$

$$\beta_1 = 1$$

$$\beta_2 = 0$$



$$\beta_0 = 1$$

$$\beta_1 = 2$$

$$\beta_2 = 1$$

CLT for Betti Numbers

- Window: $\mathbb{S}_n := [0, n] \times [0, 1]$
- Simplicial complex: $G_n := G(\mathcal{P} \cap \mathbb{S}_n, \mathcal{P}' \cap \mathbb{S}_n)$
- Betti number of dimension m in window n : $\beta_m^{(n)} := \beta_m(G_n)$

Theorem

If $\gamma < 1/4$, $\gamma' < 1/(4(m+1))$, then

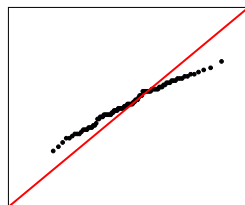
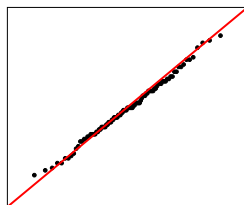
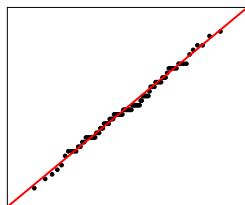
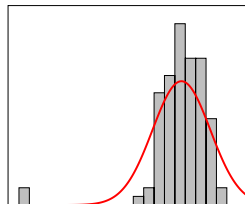
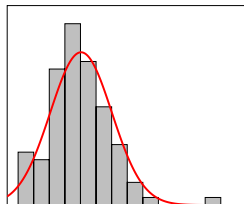
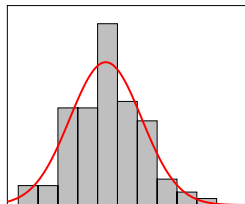
$$\frac{\beta_m^{(n)} - \mathbb{E}[\beta_m^{(n)}]}{\sqrt{\text{Var}(\beta_m^{(n)})}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Proof Idea.

Use CLT for Poisson functionals

Penrose, M. D. and Yukich, J. E. "Central limit theorems for some graphs in computational geometry." *Annals of Applied Probability* 11 (2001) □

Simulated Distribution of Betti-1 Numbers

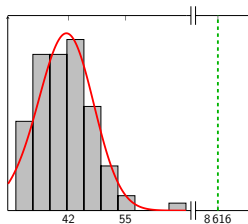


(a) $\gamma = 0.25$

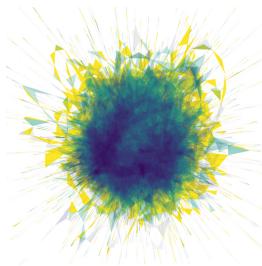
(b) $\gamma = 0.50$

(c) $\gamma = 0.75$

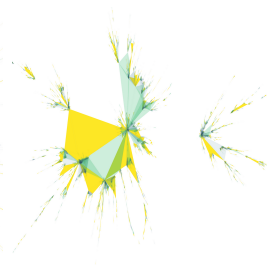
Hypothesis Test



(a) Hypothesis Test



(b) Network Data



(c) Simulation

Table of Contents

1 Introduction

2 Model

3 Results

4 Proofs

Asymptotics of Simplex Counts

- $S_n := \#\{\Delta_m \in \Sigma_m : c(\Delta_m) \in [0, n]\}$

Theorem

Central Limit Theorem

- $0 < \gamma < \frac{1}{2}; 0 < \gamma' < \frac{1}{2m+1}$
- $n^{-\frac{1}{2}}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Stable Limit Law

- $\frac{1}{2} < \gamma < 1; 0 < \gamma' < \frac{1}{2m+1}$
- $n^{-\gamma}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{S}(1/\gamma)$

Proof Sketch.

- Harris-FKG theorem
- Upper bound sum of covariance terms
- Apply CLT (associated rvs)
- Neglect high-mark vertices
- Approximate low-mark vertices as iid Pareto rvs



Stable Distribution of Simplex Counts

Theorem (Stable Limit Law)

- $X_i, i \geq 1$ iid nonnegative rvs;
- $\mathbb{P}(X_1 > x) \sim Ax^{-\alpha}$ for some $\alpha \in (1, 2)$

Then

$$n^{-1/\alpha} \sum_{i \leq n} (X_i - \mathbb{E}[X_i]) \xrightarrow{d} \mathcal{S}(\alpha)$$

Whitt, W. "Stochastic-Process Limits." *Springer Verlag* (2002)

Stable Distribution of Simplex Counts

$$\begin{aligned} \mathbb{S}_{n, \leq u} &:= [0, n] \times [0, u] \\ \mathbb{S}_{n, \geq u} &:= [0, n] \times [u, 1] \end{aligned} \quad p := (x, u) \in \mathcal{P}$$

$$\begin{aligned} d_m(p) &:= \#\{\Delta_m \in \Sigma_m : c(\Delta_m) = p\} \\ &= \frac{1}{m!} \sum_{(P_1, \dots, P_m) \in (\mathcal{P} \cap \mathbb{S}_{\geq u})^m} \mathbb{1}\{\mathcal{P}' \cap B(\{p, P_1, \dots, P_m\}) \neq \emptyset\} \end{aligned}$$

Idea: decompose S_n to low and high mark vertices:

$$u_n := n^{-b} \quad S_n = S_n^{(1)} + S_n^{\geq} := \sum_{P_i \in \mathcal{P} \cap \mathbb{S}_{n, \leq u_n}} d_m(P_i) + \sum_{P_i \in \mathcal{P} \cap \mathbb{S}_{n, \geq u_n}} d_m(P_i)$$

First Approximation – $S_n^{\geq}$ is Negligible

- Need to show: $n^{-\gamma}(S_n^{\geq} - \mathbb{E}[S_n^{\geq}]) \xrightarrow[n \uparrow \infty]{\mathbb{P}} 0$
- Chebysev's inequality

$$\begin{aligned} \text{Var}(S_n^{\geq}) &:= I + II := \lambda \int_{\mathbb{S}_{n, \geq u_n}} \mathbb{E}[\text{d}_m(p)^2] \text{d}p \\ &+ \lambda^2 \iint_{\mathbb{S}_{n, \geq u_n}^2} |\text{Cov}(\text{d}_m(p_1), \text{d}_m(p_2))| \text{d}p_1 \text{d}p_2 \end{aligned}$$

- First term:

$$I \leq C_1 n \lambda \int_{u_n}^1 u^{-2\gamma} \text{d}u \in o(n^{2\gamma}) \quad \gamma > 1/2$$

- Second term:

$$II \leq C_2 n \lambda^2 \int_{u_n}^1 u^{1-3\gamma} \text{d}u \in o(n u_n^{1-2\gamma}) \in o(n^{2\gamma}) \quad \gamma > 1/2$$

Second Approximation – Sum of Expected Degrees

Approximation with sum of expected degrees

$$\mu_m(u) := \mathbb{E}[\mathrm{d}_m(0, u)] \quad S_n^{(2)} := \sum_{(X, U) \in \mathcal{P} \cap \mathbb{S}_n, \leq u_n} \mu_m(U)$$

$$\begin{aligned} \mathbb{E}[|S_n^{(1)} - S_n^{(2)}|] &\leq n\lambda \int_0^{u_n} \mathbb{E}[|\mathrm{d}_m(0, u) - \mu_m(u)|] \mathrm{d}u \\ &\leq n\lambda \int_0^{u_n} \sqrt{\mathrm{Var}(\mathrm{d}_m(0, u))} \mathrm{d}u \end{aligned}$$

We can show $\mathrm{Var}(\mathrm{d}_m(0, u)) \in O(u^{1-3\gamma})$

$$\mathbb{E}[|S_n^{(1)} - S_n^{(2)}|] \leq nu_n^{3(1-\gamma)/2} \in o(n^\gamma)$$

Third Approximation – Sum of iid Poisson rv-s

$$N \sim \text{Poi}(\lambda n) \quad S_n^{(3)} := \sum_{i \leq \lceil \lambda n \rceil} \mu_m(U_i) \mathbb{1}\{U_i \leq u_n\}$$

$$\begin{aligned} \mathbb{E}[|S_n^{(2)} - S_n^{(3)}|] &\leq \mathbb{E}[|N - \lceil \lambda n \rceil|] \int_0^{u_n} \mu(u) \, du \\ &\leq (\sqrt{\text{Var}(N)} + 1) \underbrace{\int_0^{u_n} \mu(u) \, du}_{\in O(u_n^{1-\gamma})} \in O(\sqrt{n} u_n^{1-\gamma}). \end{aligned}$$

If $\gamma > 1/2$, then $1/2 - b(1 - \gamma) < \gamma$ so $n^{-\gamma} \mathbb{E}[|S_n^{(2)} - S_n^{(3)}|] \rightarrow 0$.
Note that

$$\sum_{i \leq \lceil \lambda n \rceil} \mu(U_i) \mathbb{1}\{U_i > u_n\} \xrightarrow[n \uparrow \infty]{L^2} 0.$$

SLT implies $n^{-1/\gamma} \sum_{i \leq \lceil \lambda n \rceil} (\mu(U_i) - \mathbb{E}[\mu(U_i)]) \xrightarrow[n \uparrow \infty]{d} \mathcal{S}(1/\gamma)$.

CLT for Poisson Functionals

Theorem (Penrose, Yukich)

Add-one cost operators:

$$\delta(G_n, o) := \beta_m(G_{n,o}) - \beta_m(G_n)$$

$$\delta(G_n, o') := \beta_m(G_{n,o'}) - \beta_m(G_n)$$

If

Moment Conditions

$$\sup_{n \geq 1} \mathbb{E}[\delta(G_n, o)^4] < \infty$$

$$\sup_{n \geq 1} \mathbb{E}[\delta(G_n, o')^4] < \infty$$

Weak Stabilization

$$\lim_{n \uparrow \infty} \delta(G_n, o) < \infty$$

$$\lim_{n \uparrow \infty} \delta(G_n, o') < \infty$$

Then the CLT holds.

Moment Condition for $\|o\|$

Remark: $\delta(G_n, o) \leq |\{\Delta_m \in \Sigma : \Delta_m \subseteq \mathbb{S}_n, o \in \Delta_m\}|$

$$\begin{aligned} \mathbb{E}[\delta(G_n, o)^4] &\leq \mathbb{E}\left[\left(\sum_{p_m \in \mathcal{P}_m^m} \mathbb{1}\{\mathcal{P}'(B(\{o\} \cup p_m)) \geq 1\}\right)^4\right] \\ &\leq \mathbb{E}\left[\left(\sum_{p' \in \mathcal{P}'} \binom{\mathcal{P}(B(p'))}{m} \mathbb{1}\{p' \in B(o)\}\right)^4\right] \end{aligned}$$

Expand the power using the multinomial theorem.

$$\begin{aligned} \mathbb{E}[\delta(G_n, o)^4] &\leq c_1 \mathbb{E}\left[\sum_{p'_4 \in \mathcal{P}'_4} \prod_{i \leq 4} \mathcal{P}(B(p'_i))^m \mathbb{1}\{p'_4 \subseteq B(o)\}\right] \\ &\quad + c_2 \mathbb{E}\left[\sum_{p'_3 \in \mathcal{P}'_3} \mathcal{P}(B(p'_1))^{2m} \prod_{i=2}^3 \mathcal{P}(B(p'_i))^m \mathbb{1}\{p'_3 \subseteq B(o)\}\right] + \dots \end{aligned}$$

Upper Bound of the First Term

Notations: $o := (0, u)$, $p'_i := (z_i, w_i)$

$$\begin{aligned} I &= c_1 \mathbb{E} \left[\sum_{\substack{p'_4 \in \mathcal{P}'^4 \\ \neq}} \prod_{i \leq 4} \mathcal{P}(B(p'_i))^m \mathbb{1}_{\{p'_4 \subseteq B(o)\}} \right] \\ &= c_1 \iint_{[0,1] \times \mathbb{S}^4} \mathbb{E} \left[\prod_{i \leq 4} \mathcal{P}(B(p'_i))^m \mathbb{1}_{\{p'_i \in B((0, u))\}} \right] dp'_4 du \end{aligned}$$

Indicator: $\mathbb{1}_{\{p'_i \in B((0, u))\}} = \mathbb{1}_{\{|z_i| \leq u^{-\gamma} w_i^{-\gamma'}\}}$ Upper bound with Hölder's inequality

$$I \leq 2^4 c_1 \iint_{[0,1] \times [0,1]^4} \prod_{i \leq 4} \left(\int_0^{\beta u^{-\gamma} w_i^{-\gamma'}} \mathbb{E} \left[\mathcal{P}(B((z_i, w_i)))^{4m} \right]^{1/4} dz_i \right) du dw$$

CLT for Poisson Functionals

Theorem (Penrose, Yukich)

Add-one cost operators:

$$\delta(G_n, o) := \beta_m(G_{n,o}) - \beta_m(G_n)$$

$$\delta(G_n, o') := \beta_m(G_{n,o'}) - \beta_m(G_n)$$

If

Moment Conditions

$$\begin{aligned} \sup_{n \geq 1} \mathbb{E}[\delta(G_n, o)^4] &< \infty \\ \sup_{n \geq 1} \mathbb{E}[\delta(G_n, o')^4] &< \infty \end{aligned}$$

Weak Stabilization

$$\begin{aligned} \lim_{n \uparrow \infty} \delta(G_n, o) &< \infty \\ \lim_{n \uparrow \infty} \delta(G_n, o') &< \infty \end{aligned}$$

Then the CLT holds.

Weak Stabilization for " $\mathcal{O}$ "

"holes = cycles - boundaries":

$$\beta_m(G_n) = \dim(Z(G_n)) - \dim(B(G_n))$$

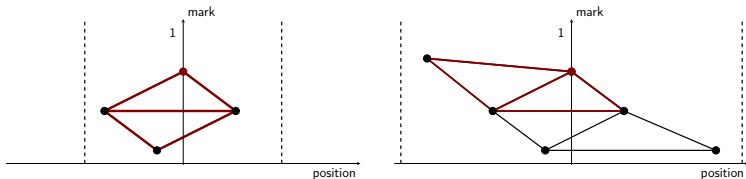
Idea: show weak stabilization for **cycles** and boundaries separately.

- Boundedness: $\dim(Z_{n,\mathcal{O}}) - \dim(Z_n) \leq \deg_{m,n}(\mathcal{O}) < \infty$
- Monotonicity (need to show)

Monotonicity

- Let $n_1 \leq n_2$, and consider the canonical map

$$Z_{n_1,o} \rightarrow Z_{n_2,o}/Z_{n_2}$$



- Claim: $\text{kernel } Z_{n_1,o} \cap Z_{n_2} = Z_{n_1}$
 - $\Delta_m \in \Sigma_{n_1} \implies \Delta_m \in \Sigma_{n_1,o} \cap \Sigma_{n_2} \implies Z_{n_1} \subseteq Z_{n_1,o} \cap Z_{n_2}$
 - $Z_{n_1} \supseteq Z_{n_1,o} \cap Z_{n_2}$
- Induced map: $Z_{n_1,o}/Z_{n_1} \rightarrow Z_{n_2,o}/Z_{n_2}$ is injective
- $\dim(Z_{n_1,o}) - \dim(Z_{n_1}) \leq \dim(Z_{n_2,o}) - \dim(Z_{n_2})$

- Motivation
 - Create higher-order network models for complex systems
- Model
 - Random Connection Hypergraph Model (RCHM)
 - Stochastic model based on Poisson point processes
 - Simplicial Complex: representation of hyperedges
- Properties
 - Degree distribution
 - Simplex counts
 - Betti numbers

Thank you for your attention!

References



A.-L. Barabási and R. Albert, *Emergence of scaling in random networks*, Science **286** (1999), no. 5439, 509–512.



P. Gracar, A. Grauer, L. Lühtrath, and P. Mörters, *The age-dependent random connection model*, Queueing Systems **93** (2019), no. 3, 309–331.



C. Hirsch and P. Juhász, *On the topology of higher-order age-dependent random connection models*, arXiv preprint arXiv:2309.11407 (2023).



G. Last and M. D. Penrose, *Lectures on the Poisson process*, vol. 7, Cambridge University Press, 2017.



M. D. Penrose and J. E. Yukich, *Central limit theorems for some graphs in computational geometry*, Annals of Applied probability **11** (2001), no. 4, 1005–1041.



W. Whitt, *Stochastic-process limits*, Springer New York, NY, 2002.